

Derek Chen

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EDUCATION

The Cooper Union for the Advancement of Science and Art

Bachelor of Science in Computer Science, GPA: 3.816

New York, NY

Aug. 2025 – Current

EXPERIENCE

Fellow

Spring 2026

CodePath

Remote

- Selected for CodePath's Intermediate Web Development fellowship, a rigorous industry-mentored program covering full-stack web fundamentals
- Built and deployed full-stack web applications by applying advanced front-end and back-end concepts under weekly technical assessments
- Sharpened technical interview readiness through CodePath's TIP102 curriculum, solving data structures and algorithms problems in Python across weekly timed practice sessions
- Collaborated with industry professional mentors and peers to review code and ship features on a team-based project timeline

Fellow

Summer 2025

All Star Code

Remote

- Earned a seat in a competitive, application-only fellowship by demonstrating strong fundamentals in computer science, entrepreneurship, and professional development
- Built full-stack web applications with peers and mentors using HTML, CSS, JavaScript, and Firebase, shipping features from design through deployment
- Integrated third-party APIs and managed Firebase databases to support real-time data storage and retrieval across projects

Intern

Summer 2023

UA3

New York, NY

- Distributed food to over 1,500 community members weekly by coordinating pantry logistics and volunteer efforts over a six-week period
- Maintained daily pantry and office operations, ensuring consistent service delivery for the community over the course of the internship
- Increased civic participation by engaging community members in voter registration campaigns during weekly distribution events

PROJECTS

PengPlayer | C++, Spotify Web API, librespot, MSVC, vcpkg

2025 – Present

- Built a native Windows terminal-based (TUI) Spotify player in C++ by implementing OAuth Authorization Code Flow with automatic token persistence and refresh
- Integrated librespot for audio playback by managing the subprocess via the Windows API and polling for device readiness before playback
- Designed a REST client layer with cpp-httplib and nlohmann/json to wrap the Spotify Web API, supporting playback control and queue management

CO2 in Manhattan Data Viz | Python, NumPy, Pandas, Matplotlib, Seaborn

Fall 2025

- Visualized building-level carbon emissions across Manhattan by building a heatmap from the NYC Local Law 84 open dataset
- Cleaned and processed raw emissions data using Pandas and NumPy to prepare it for geographic visualization
- Presented findings using Matplotlib and Seaborn to highlight emissions hotspots across the borough

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, HTML, CSS

Frameworks & Libraries: React, P5.js

Data & Visualization: NumPy, Pandas, Matplotlib, Seaborn

Platforms & Tools: Firebase, REST APIs, Git/GitHub